

Sessional 1 Exam Topics
General Introduction + Introduction to Data structures+ Importance of course for program and contents to be covered in the course + General Programming Concepts
General Programming Concepts that include Abstract Data Types+ pointers+ arrays+ Solving examples
Programming Concepts that include structures + Accessing structure elements with dot operator and arrow operator + solving examples+ Row Major and Column Major Arrangements of Elements from Memory Representation point of view+ Pointers and Arrays, Pointers with Multi-Dimensional Arrays, Arrays of Pointers, Pointers and Strings, Array of Pointers to String
Complexity, Big O, Omega and theta. Dynamic Memory Allocation using New command.
Linked List, Singly Linked Lists and operations on Singly Linked Lists i.e. List creation, Traversal of Linked List (At front and at end)
Searching, Insertion, Deletion, Display of elements, Reversal of Linked List elements etc.
Stack handling, Push() and Pop() Operations on Stack, Overflow state and underflow state. Implementation with arrays and linklist. Application and significance of Stack with daily life examples+ Brackets validation with stacks.
Reversal of String, Checking Validity of An expression containing nested parentheses Polish Notations with Arithmetic Expression+ Converting Infix Expression into Post Fix and pre fix Notation
Evaluation of Post Fix, Pre fix and Infix Expressions+ Converting infix equation to postfix
Introduction to recursion+ Recursive chain + Solving examples like Fibonacci series and binary searching algorithm with recursion.